Lab 4 – SDN Controller on Packet Tracer

Student Full Name _Greemesh Basnet___ _________/10 Marks

Objective

To use a SDN Controller to gather information about a network and configure network settings.

Background

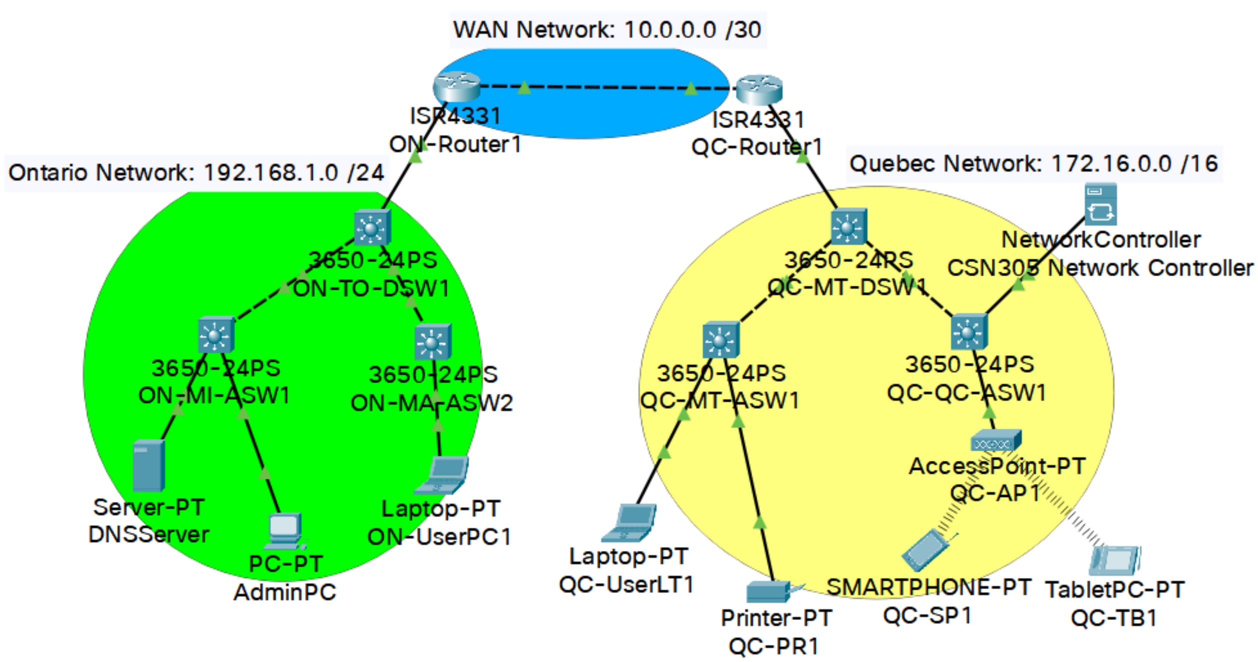
- Use the material posted from weeks 4 and 5 to identify the architecture of a Network Controller.
- Use your previous knowledge of IP addressing, Subnetting, securing routers and switches access, and SSH protocol to set a complete network with two LAN sites separated by a WAN network. The network will be controllable from a SDN Network controller.
- SDN requires network devices that are to be controlled, to run a software version higher or equal to **Version 16**. Packet Tracer **ISR4331 routers and 3650-24PS switches** meet that condition. Use them for the topology of this lab.

Instructions

1. Use a different font and color to write your answers.
2. Do not erase anything from this document.
3. Resize the screenshots to make them easily readable.
4. Refer to the Appendix A to check your Packet Tracer settings.

Part A: Preparing the Network to be Controllable. (3 marks out of 10)

1. Adjust the Packet Tracer preferences settings as shown in the Appendix A.
2. Create a topology as shown in the following figure below. All the routing is made based on RIP. All the routers and switches are SSH-secured. **Be sure that you update the addresses according to the address table in the next page.**



Use the following address table: replace the XX characters with the last two digits of your student ID **[One mark]**

Device	Interface	Address / Mask	Gateway
ON-Router 1	GigabitEthernet 0/0/0	10.0.0.1 /30	
ON-Router 1	GigabitEthernet 0/0/1	192.168.23.1 /24	
ON-TO-DSW1	VLAN 1	192.168.23.2 / 24	192.168.23.1
ON-TO-ASW1	VLAN 1	192.168.23.3 / 24	192.168.23.1
ON-MA-ASW2	VLAN 1	192.168.23.4 / 24	192.168.23.1
DNSSever	FastEthernet0	192.168.23.254	192.168.23.1
AdminPC	FastEthernet0	192.168.23.10	192.168.23.1
ON-UserPC1	FastEthernet0	192.168.23.11	192.168.23.1
QC-Router 1	GigabitEthernet 0/0/0	10.0.0.2 /30	
QC-Router 1	GigabitEthernet 0/0/1	172.16.23.1 /16	
QC-MT-DSW1	VLAN 1	172.16.23.2 / 16	172.16.23.1
QC-MT-ASW1	VLAN 1	172.16.23.3 / 16	172.16.23.1
QC-QC-ASW1	VLAN 1	172.16.23.4 / 16	172.16.23.1
QC-UserLT1	FastEthernet0	172.16.23.10 / 16	172.16.23.1
QC-PR1	FastEthernet0	172.16.23.11 / 16	172.16.23.1
QC-SP1	Wireless0	172.16.23.12 / 16	172.16.23.1
QC-TB1	Wireless0	172.16.23.13 / 16	172.16.23.1
CSN305 Network Controller	GigabitEthernet0	172.16.23.254 / 16	172.16.23.1

Other configurations to be done:

1. DNS server address 192.168.XX.254 is set on all the end devices. Domain name is "**csn305.ca".**
2. All the switches and routers devices names are set as indicated in this table.
3. **Basic configuration of the six switches and two routers** (be sure that you use the **enable** mode and **configure terminal** commands).

   To ensure remote IP access to the switch, remember to use the following commands in each switch:

   (config)# ip default-gateway <*IP address of the default gateway*>

   (config)# interface vlan 1

   (config-if)# ip address <*IP address*> <*subnet mask*>

   (config-if)# no shutdown

Then, you can continue the configuration mode to enter the basic security settings:

hostname *(use the name given in the table above)*

ip domain-name **csn305.ca**

crypto key generate rsa

*(enter the value **2048** after the IOS prompt request)*

enable password *(enter your own password including at least 8 characters, with at least one capital letter, one symbol, and one number)*

line console 0

password *(enter your own password including at least 8 characters, with at least one capital letter, one symbol, and one number)*

line vty 0 4

transport input ssh

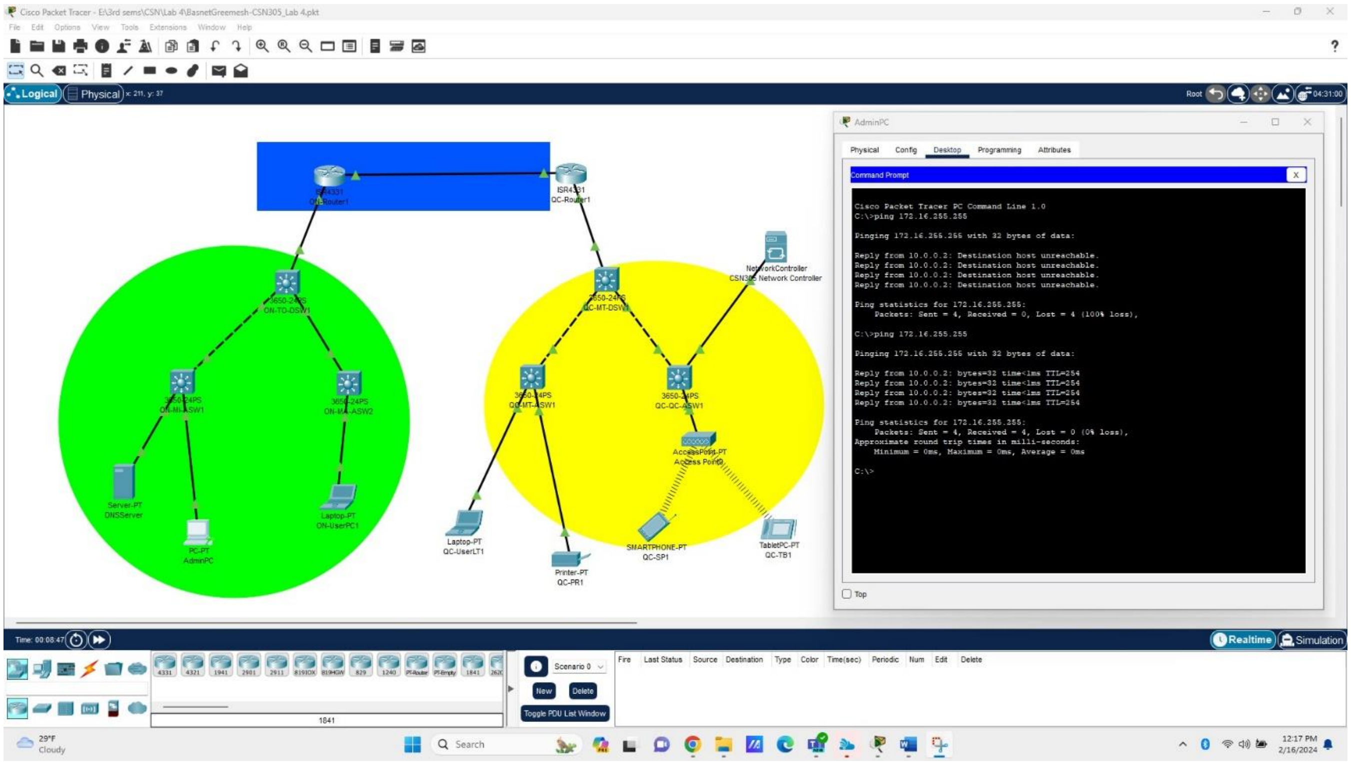
login local

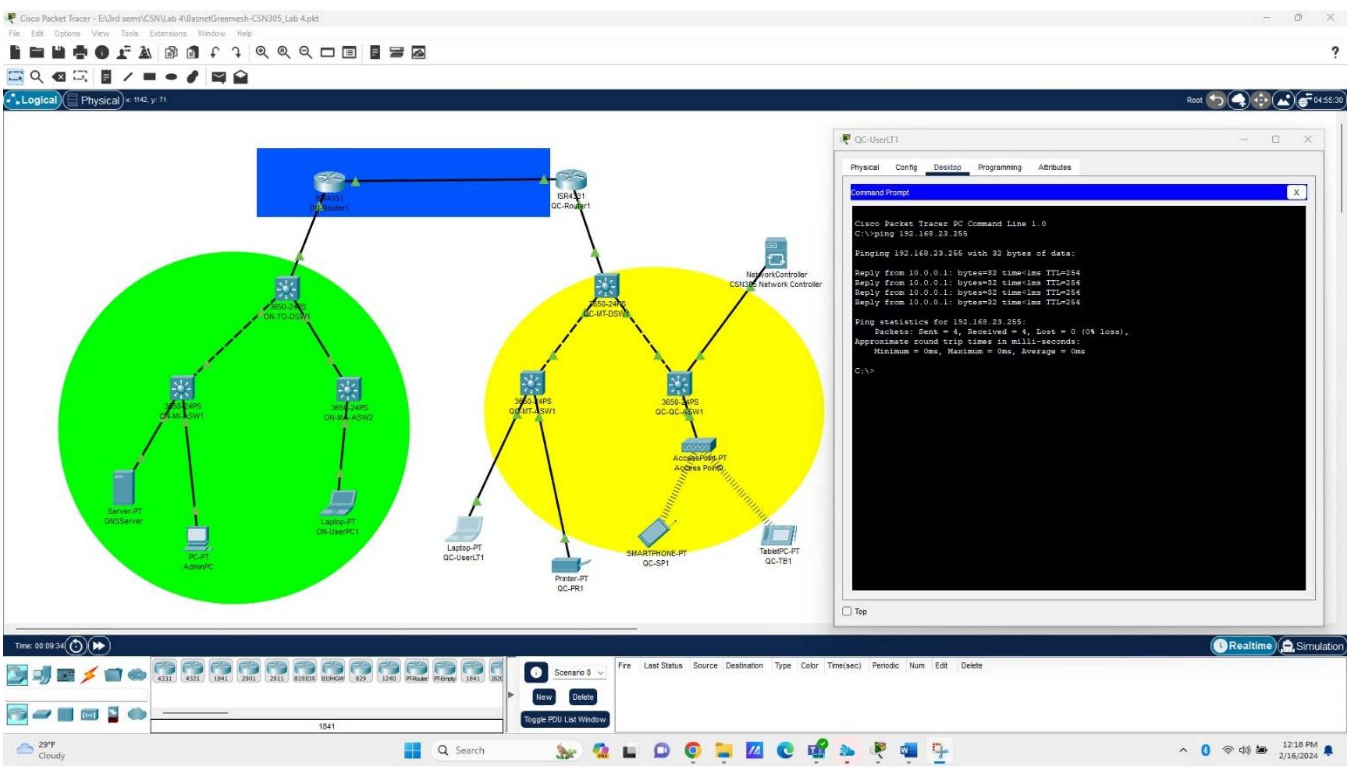
username **csn305** password *(enter your own password including at least 8 characters, with at least one capital letter, one symbol, and one number)*

exit

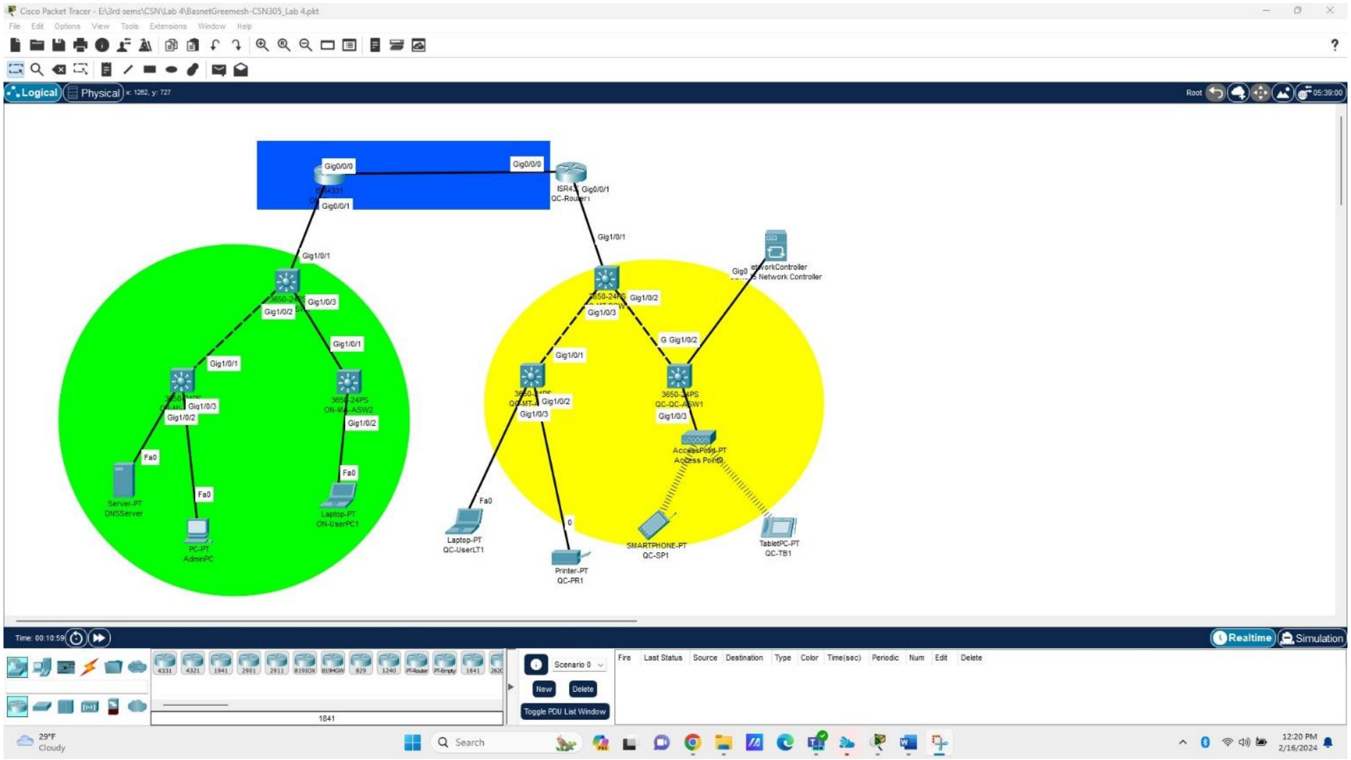
4. Once you complete the configuration of each device, exit to the user mode, and check if the password is required now to access the enable mode.
5. **Routing settings:** all the routers do RIP routing. Be sure that you advertise all the three networks (10.0.0.0 /30, 172.16.XX.0 /16 and, 192.168.XX.0 /24) on each router to ensure full connectivity end-to-end. Verify it by issuing ping commands on each network segment as needed. Troubleshoot if necessary.
6. **End devices settings:** be sure that you configure each end device with its corresponding interface IP address as well as the gateway address as indicated in the address table above.

7. **Ping**: use the command **ping** <broadcast address of the remote network>**.** Insert screenshots of successful ping commands from one Ontario end device and one Quebec end device here. Be sure that the destination address is the broadcast address of the other network. **[One mark for each ping]**



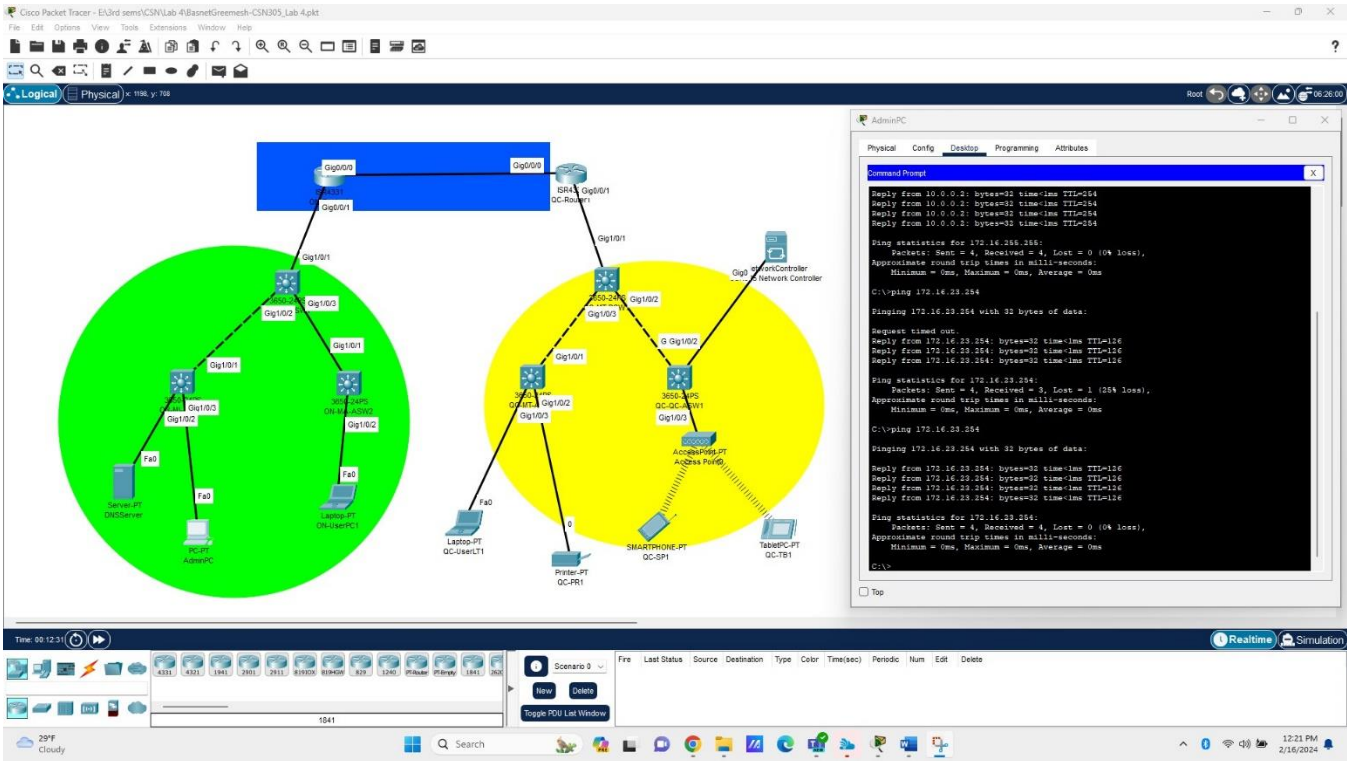


8. Insert your Packet Tracer topology screenshot here. **[One mark]**



Part B: Setting and verifying a Network Controller (4 marks out of 10)

1. Verify if the controller is accessible from the Admin-PC by sending a ping command. Insert a screenshot of successful ping command from Admin-PC to Network Controller here. **[One mark]**



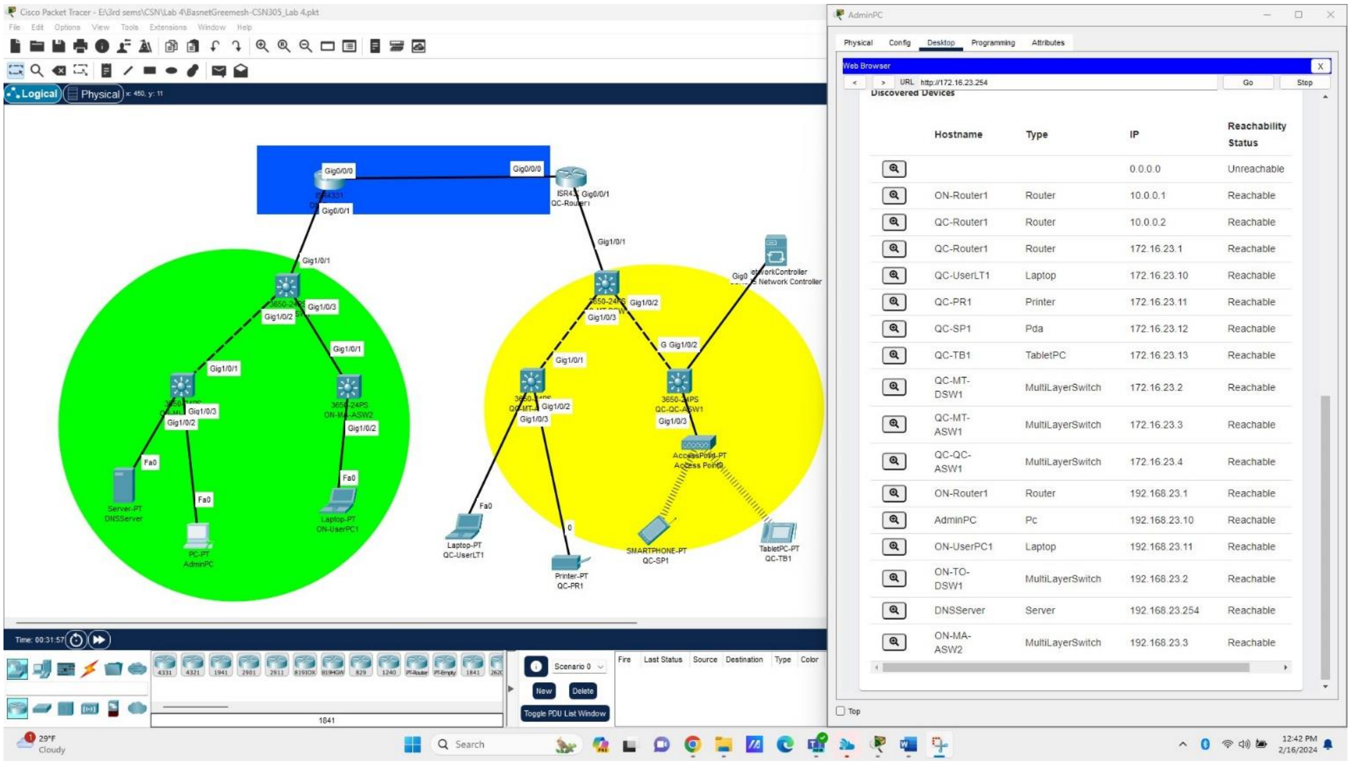
2. Open the Network Controller **Config** tab. On the left, under REAL WORLD, click Controller. If the Server Status is Stopped, move on to the next sub-step. If the Server Status is Disabled in Preferences, then you will need to enable external access by following these instructions:

   i) Got to the main Packet Tracer menu and select **Options > Preferences**. Click the **Miscellaneous** tab., and under **External Network Access**, click **Enable External Access for Network Controller REST API**. Close it.

   ii) Return to the Network Controller **Config** and on the left under **REAL WORLD**, click **Controller**. The Server Status should now be Stopped. Click **Access Enabled** to enable it. Server Status changes to "**Listening on port 58000**". If the port is some other value, change it to 58000. This is the port number in the Python scripts.

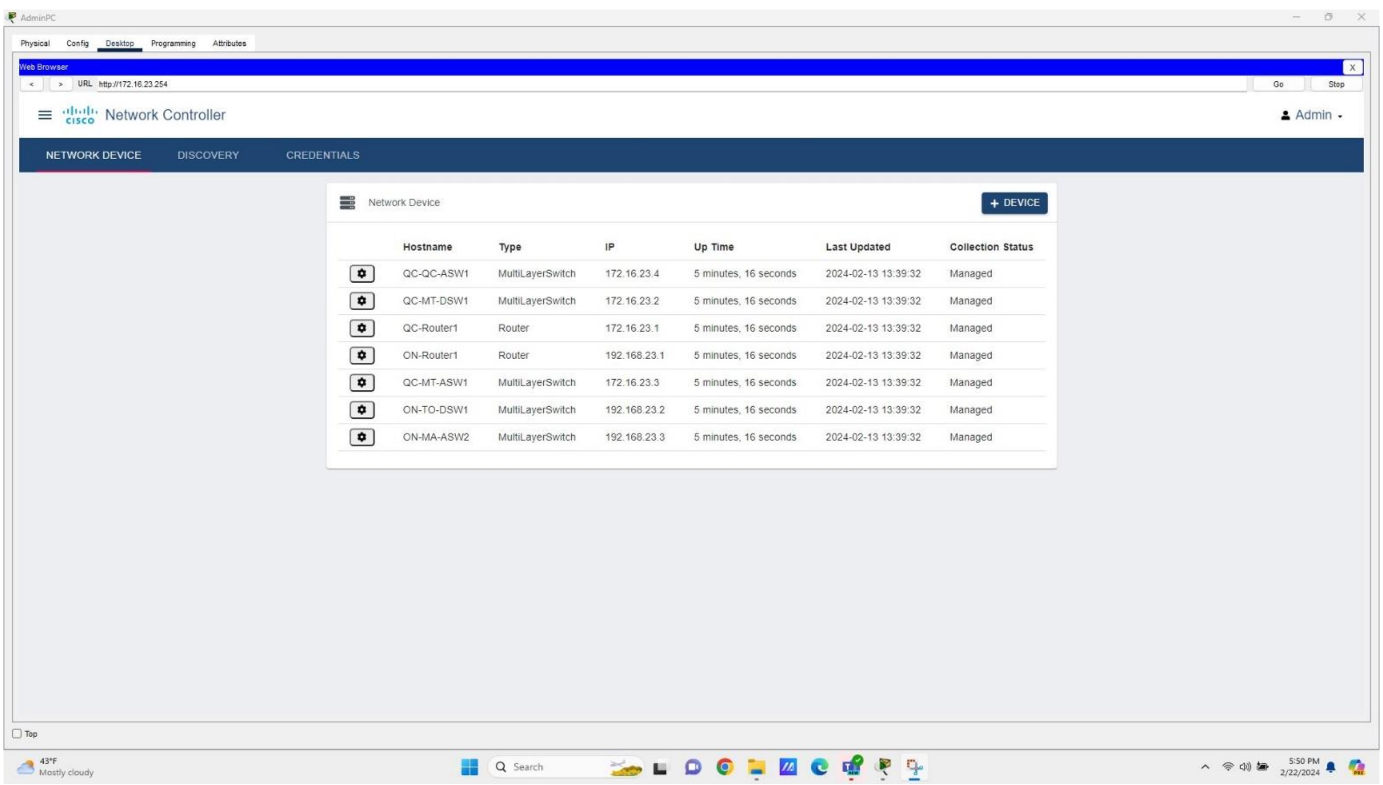
3. Open the web browser from Admin-PC and access the IP address of the network Controller. Create the **Admin** username and (for the sake of simplicity, not as an advisable security practice) use the same password as part A. Do login. At the beginning all the counters shown should be in 0. Go to the **Credentials** menu and set the same username and password as you set the SSH protocol in all the networking devices (routers and switches). Use **CSN305 Network Admin** as description text.

4. Go to the **Discovery** menu and set a new discovery procedure. The Type is set to **CDP**. The name to **CSN305 Network**, the IP address is the same IP address of the VLAN 1 where the Network Controller is directly connected (172.16.XX.4). the credential is the same created in the previous step. Leave the other options by default. Wait for the Cisco Discovery Protocol (CDP) to progress and click on the Network name once its status is "**Terminated**". Insert the screenshot of all the **Discovered Devices** list here. **[One mark]**



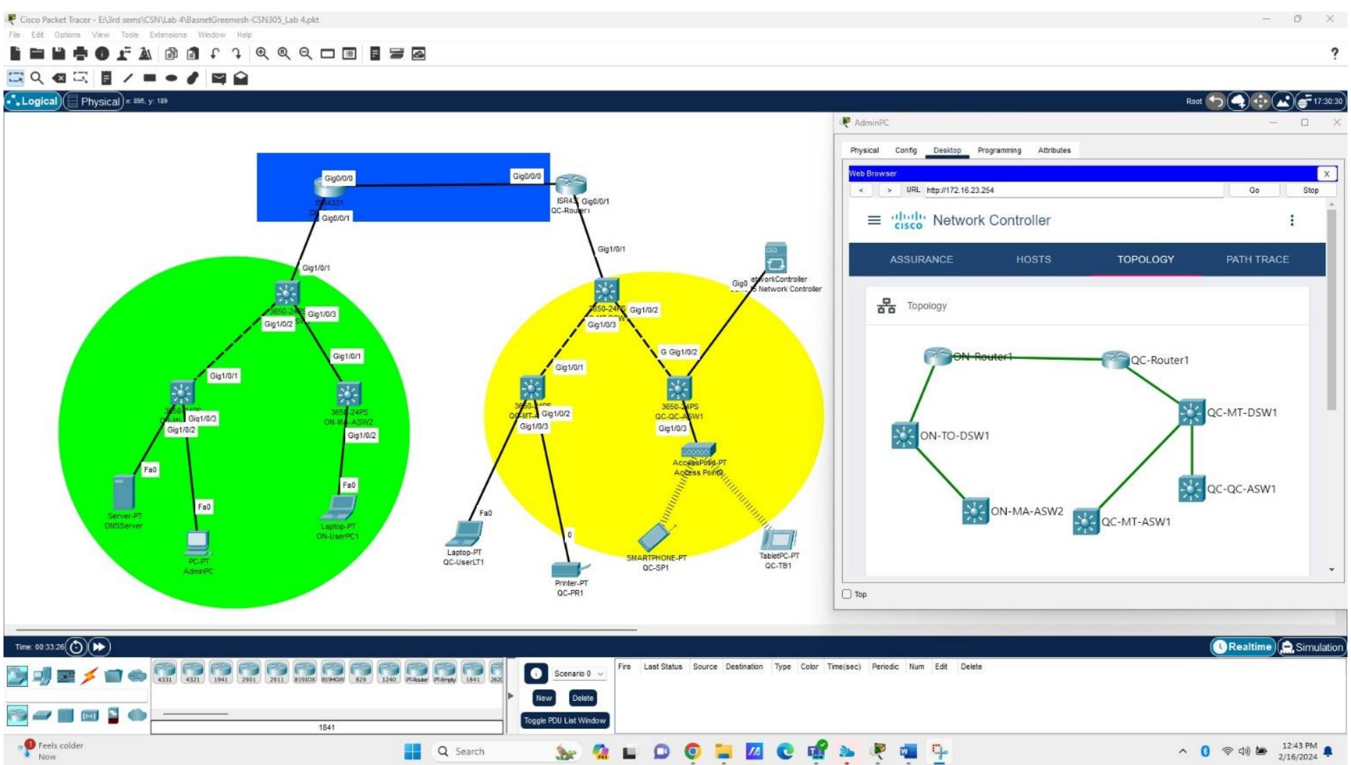
	Hostname	Type	IP	Reachability Status
			0.0.0.0	Unreachable
	ON-Router1	Router	10.0.0.1	Reachable
	QC-Router1	Router	10.0.0.2	Reachable
	QC-Router1	Router	172.16.23.1	Reachable
	QC-UserLT1	Laptop	172.16.23.10	Reachable
	QC-PR1	Printer	172.16.23.11	Reachable
	QC-SP1	Pda	172.16.23.12	Reachable
	QC-TB1	TabletPC	172.16.23.13	Reachable
	QC-MT-DSW1	MultiLayerSwitch	172.16.23.2	Reachable
	QC-MT-ASW1	MultiLayerSwitch	172.16.23.3	Reachable
	QC-QC-ASW1	MultiLayerSwitch	172.16.23.4	Reachable
	ON-Router1	Router	192.168.23.1	Reachable
	AdminPC	Pc	192.168.23.10	Reachable
	ON-UserPC1	Laptop	192.168.23.11	Reachable
	ON-TO-DSW1	MultiLayerSwitch	192.168.23.2	Reachable
	DNSServer	Server	192.168.23.254	Reachable
	ON-MA-ASW2	MultiLayerSwitch	192.168.23.3	Reachable

5. Go to the N**etwork Device** menu. Insert the screenshot of all the **Network Devices** list here. **[One mark]**



	Hostname	Type	IP	Up Time	Last Updated	Collection Status
	QC-QC-ASW1	MultiLayerSwitch	172.16.23.4	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	QC-MT-DSW1	MultiLayerSwitch	172.16.23.2	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	QC-Router1	Router	172.16.23.1	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	ON-Router1	Router	192.168.23.1	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	QC-MT-ASW1	MultiLayerSwitch	172.16.23.3	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	ON-TO-DSW1	MultiLayerSwitch	192.168.23.2	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed
	ON-MA-ASW2	MultiLayerSwitch	192.168.23.3	5 minutes, 16 seconds	2024-02-13 13:39:32	Managed

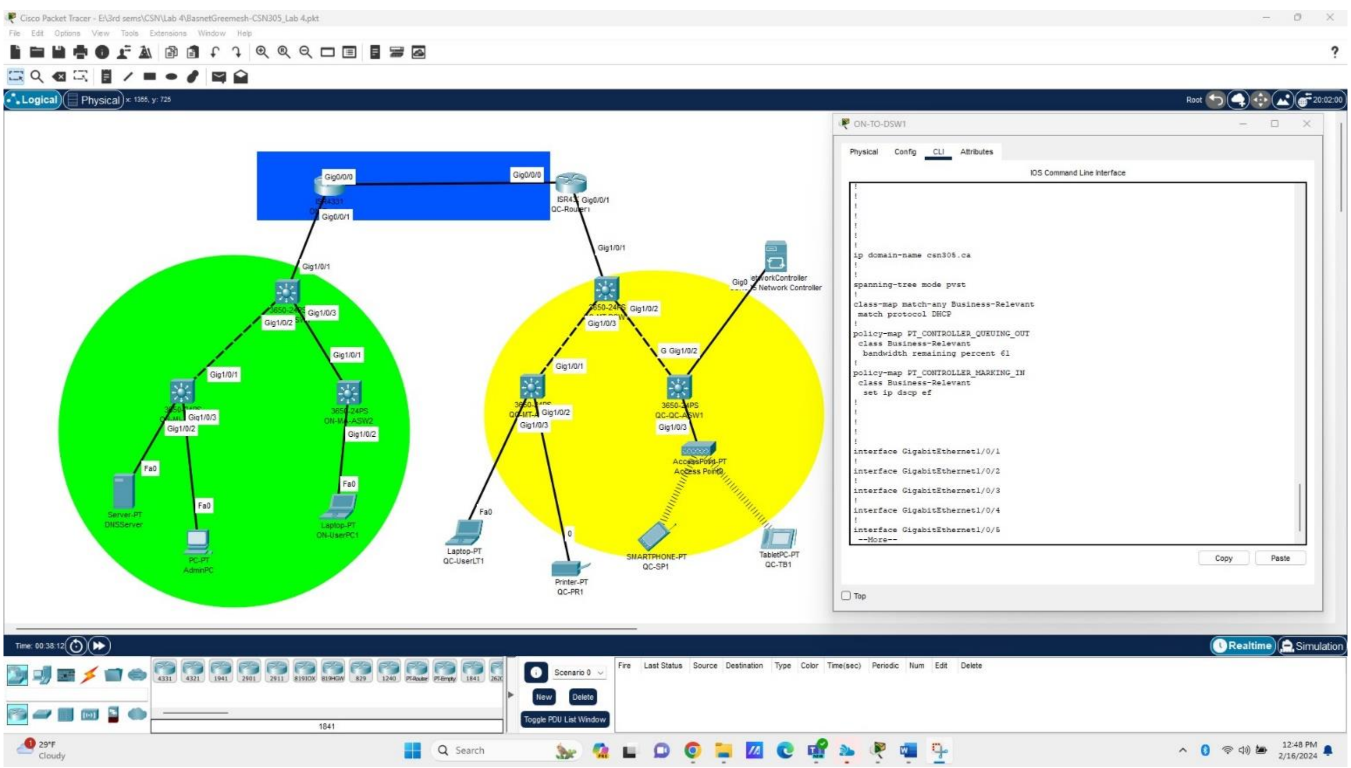
Go to the main left menu and open the **Assurance** information window. Click on **Topology.** You can drag the icons to make them easily readable. Insert the screenshot of the **discovered topology** here. **[One mark]**



Part C: Pushing configuration settings onto the managed devices (3 marks out of 10)

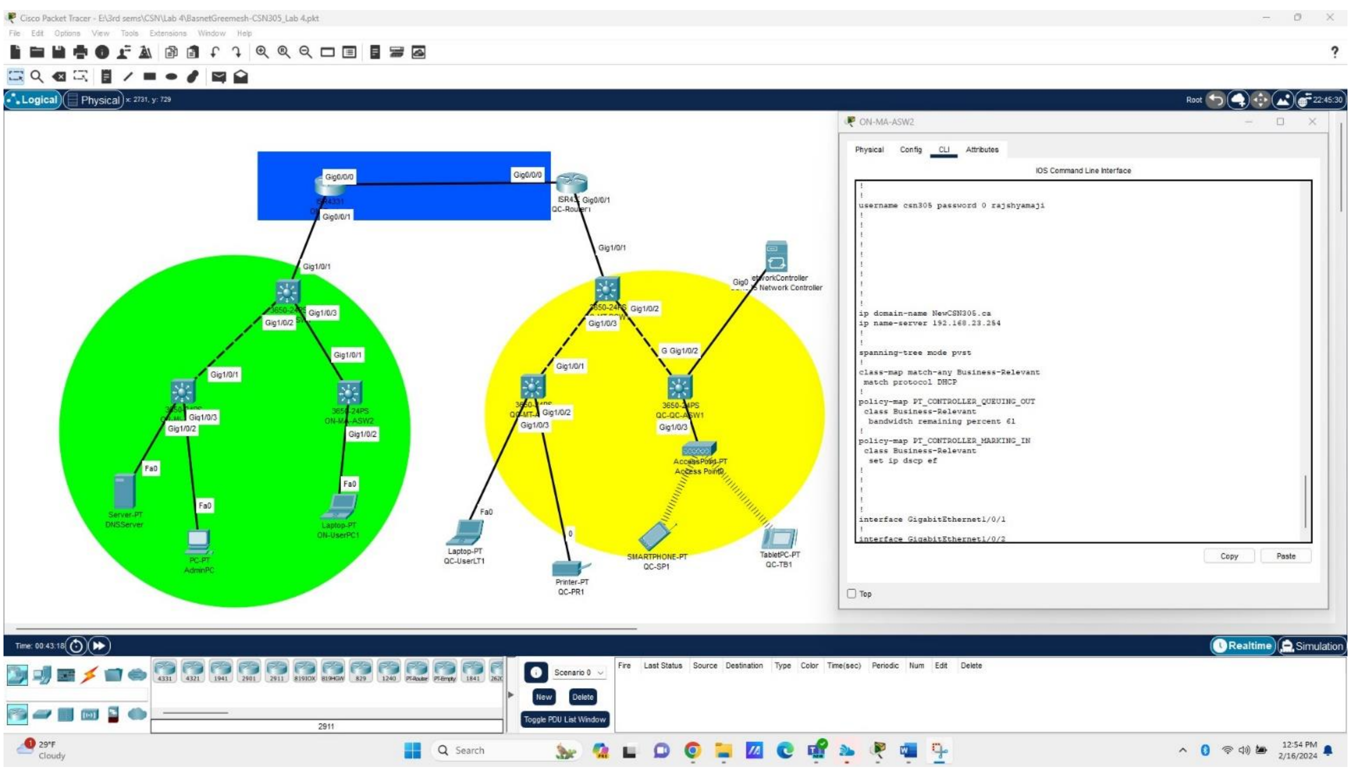
1. Go to the main left menu and select **Policy**. Create a new **Scope** named **CSN305_DHCP**. The policy will be set as **Business-relevant**, and the protocol to be selected is **DHCP**. Select the already created scope for this policy. The name of the policy will be **DHCP_Policy**.
2. Before to apply the policy to network devices editing the scope first: select all the existing switches from the network devices list that the scope will affect.
3. Now edit the DHCP_policy using the gear icon on the left side. Click on the **Apply** button.

4. **Verify applied policies on network devices:** go to any of the switches included in the DHCP_Policy. Open the CLI interface and do **show running-config**. Scroll on the output of this command to find the QoS policy applied on it from the Network Controller. Insert the screenshot of the running configuration section that shows the applied policy here (it normally goes before the interfaces list). **[One mark]**

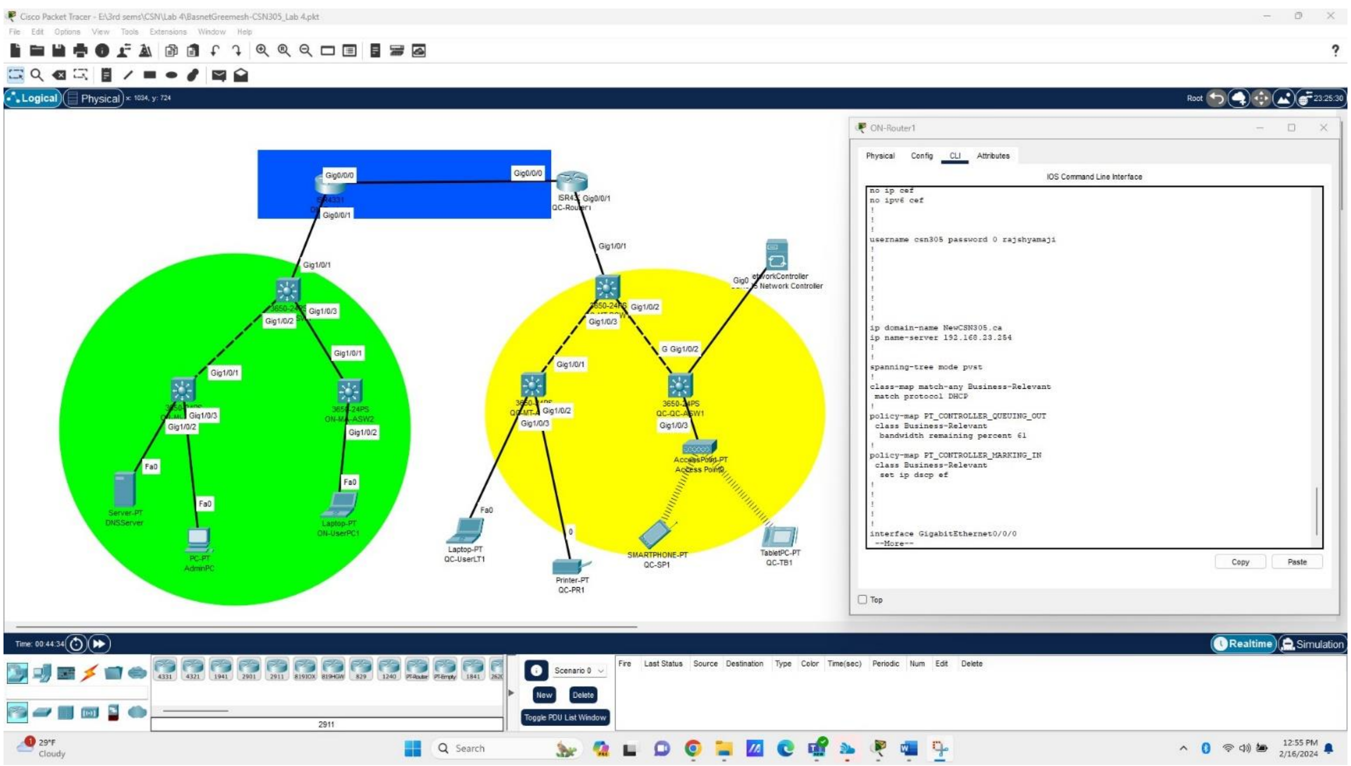


5. **Create new network settings:** go to the **Network Settings** section and create a new DNS. The domain name will be NewCSN305.ca, and the IP address is to be set to 192.168.XX.254 (the same DNS server address). Then create a new NTP server on the same server address (192.168.XX.254). Click on **Push Config** to upload the new configuration settings to your network devices.

6. Go to one of the switches in Toronto. Open the CLI interface and do **show running-config**. Scroll on the output of this command to find the QoS policy applied on it from the Network Controller. Insert the screenshots of the lines in the running configuration that show the applied policies here. **[One mark]**



7. Go to one of the routers. Open the CLI interface and do **show running-config**. Scroll on the output of this command to find the QoS policy applied on it from the Network Controller. Insert the screenshots of the lines in the running configuration that show the applied policies here. **[One mark]**



To submit:

1. Be sure that you wrote your full name at the beginning of this document. (10% deducted if you do not write your name)

2. Two files to be submitted in the same submission attempt. (One mark deducted if you do not submit both files at the same attempt)

   i) Rename this report file as **LastNameFirstName_CSN305Lab4** and save it as PDF. Upload this document including all your answers and screenshots. (10% deducted if this file is missed in your submission or wrong name is used)

   ii) Rename the Packet Tracer file as **LastNameFirstName_CSN305Lab4**. Upload this document in the same submission attempt. (10% deducted if this file is missed in your submission or wrong name is used)

Appendix A: Packet Tracer preferences settings.

Be sure that you use these settings from the beginning. **[One mark deducted if you do not use this settings]**